

North Carolina A&T State University

Department of Mechanical Engineering

MEEN 424

Vibrations & Control

Session 2 Lecture Notes

Control System Background

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These notes are prepared for students enrolled in MEEN 424.
They supplement, but do not replace, attendance and active participation.

1. Control System Definition

A **control system** consists of subsystems and processes (or **plants**) assembled for the purpose of obtaining a **desired output** with desired **performance**, given a specified **input**.

Every control system has three fundamental components:

- **Input** $r(t)$: The desired output value — what we want the system to achieve (also called the reference or setpoint).
- **Process** $G(s)$: The physical system being controlled — a motor, furnace, elevator, or robot arm.
- **Output** $y(t)$: What the system actually produces — position, temperature, speed, or angle.

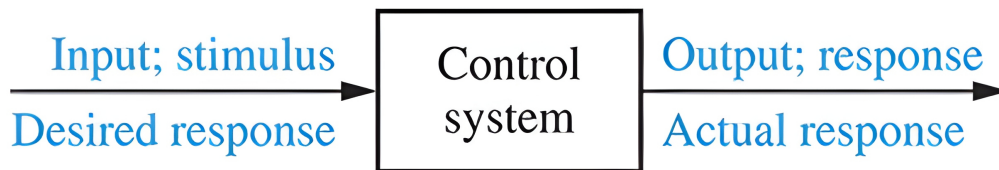


Figure 1: Simplified description of a control system

2. Motivating Example — The Elevator

When the 4th floor button is pressed from the 1st floor, the elevator rises with a speed and floor-leveling accuracy designed for passenger comfort. The button press is the **input** (desired output), and the elevator position is the **output**. Two performance measures are immediately apparent:

- **Transient response:** How the elevator moves from floor 1 to floor 4. Too fast causes passenger discomfort; too slow causes impatience. It also has structural safety implications.
- **Steady-state error:** Whether the elevator levels exactly at the 4th floor. Poor leveling compromises passenger safety and convenience. The difference between the desired floor level and the actual resting position.

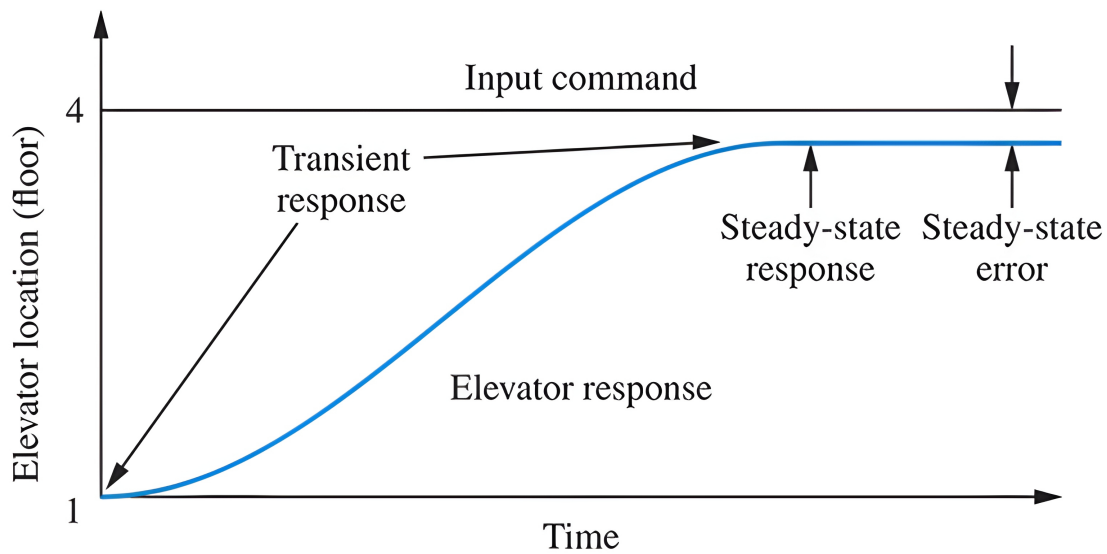


Figure 2: Elevator response to pressing the 4th floor button, showing transient response and steady-state error

3. Why We Build Control Systems

There are four primary reasons engineers design and implement control systems:

1. Power Amplification

A small input signal controls a large output force. The controller amplifies the signal to drive a powerful actuator without requiring large physical effort at the input.

Example: A steering wheel input (small torque) controls the heavy hydraulic steering system of a truck.

2. Remote Control

Control systems allow operation of plants from a distance without being physically present. The input and output can be separated by very large distances.

Example: Spacecraft attitude control from Earth; remote robotic surgery.

3. Convenience of Input Form

The input can be in a different physical form than the output. Transducers convert between forms, allowing a convenient input to command a different type of output.

Example: A thermostat dial position (mechanical) controls a furnace gas valve (electrical/thermal).

4. Compensation for Disturbances

Control systems automatically correct for unexpected disturbances that affect the

plant output — something open-loop systems fundamentally cannot do.

Example: Cruise control maintaining vehicle speed despite uphill grades and headwinds.

4. Open-Loop Systems

An **open-loop system** starts with an **input transducer** which converts the form of the input to that used by the controller. The controller then drives the plant. There is no feedback — the output has no effect on the input.

Disturbances enter via summing junctions at the plant, but there is no mechanism to detect or correct their effect on the output.

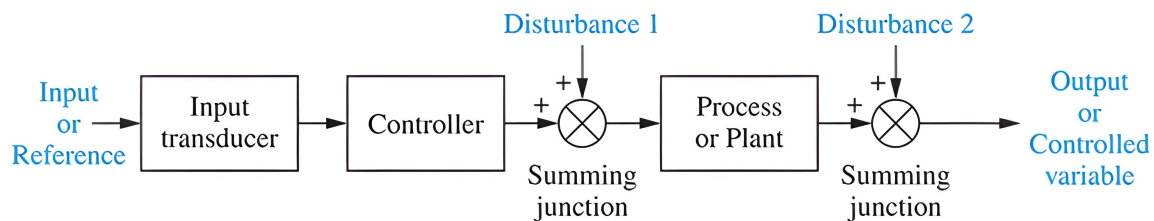


Figure 3: Open-loop control system block diagram showing input transducer, controller, plant, and disturbance

Advantages and Disadvantages

Advantages	Disadvantages
Simple and inexpensive to build	Cannot detect or correct disturbances
Easy to design — no stability analysis needed	Sensitive to plant parameter changes
Works well when plant behavior is very predictable	No accuracy guarantee on the output

5. Closed-Loop (Feedback) Systems

A **closed-loop system** uses an **output transducer** (sensor) to measure the actual output and feed it back for comparison with the reference input. If a difference (error) exists, the controller generates an **actuating signal** to drive the plant and reduce the error to zero.

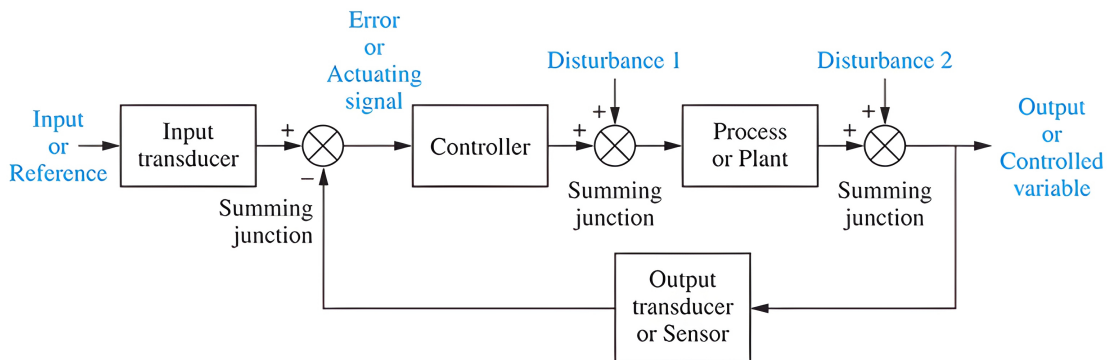


Figure 4: Closed-loop (feedback) control system block diagram showing input transducer, controller, plant, output transducer, and feedback path

The Error Signal

$$e(t) = r(t) - y(t)$$

$r(t)$ = reference input (desired output) $y(t)$ = measured output (actual output)

If $e(t) \neq 0$, the controller drives the plant to reduce the error. If $e(t) = 0$, the plant output already equals the desired value and no correction is needed.

Key Advantages Over Open-Loop

- Greater accuracy than open-loop systems
- Less sensitive to noise, disturbances, and environmental changes
- Transient response and steady-state error can be controlled more conveniently

- Performance can often be improved by simple gain adjustment in the feedback loop

6. Analysis and Design Objectives

Every control system must be analyzed and designed to meet three core objectives. These objectives are interconnected — stability must be achieved first before the others can be addressed.

1. Transient Response

Describes how the system behaves from its initial state until it reaches the final steady state. The transient response must be designed carefully:

- Too fast $\Rightarrow$ discomfort, overshoot, or structural damage
- Too slow $\Rightarrow$ impatience or delayed operation

Measured by: *rise time, settling time, percent overshoot, peak time.*

2. Steady-State Accuracy

Describes how close the final output is to the desired value after all transients have died out. Poor steady-state accuracy means the system never quite reaches its goal. Measured by: *steady-state error* $e_{ss} = \lim_{t \rightarrow \infty} e(t)$.

3. Stability

The most fundamental requirement. A system must be stable before any other design objectives are meaningful. An unstable system grows without bound and can self-destruct.

7. Stability

Total Response

$$\text{Total Response} = \underbrace{\text{Natural Response}}_{\text{system dynamics}} + \underbrace{\text{Forced Response}}_{\text{due to input}}$$

- **Natural response:** Describes how the system dissipates or acquires energy. Depends only on the system, NOT on the input. This is what determines stability.

- **Forced response:** Depends on the input signal. For a stable system, the forced response remains after the natural response decays to zero.

Stability Definitions

Condition	Description
Stable	Natural response decays to zero as $t \rightarrow \infty$. Transient dies out, leaving only the forced response. All closed-loop poles in the left half s-plane ($\text{Re}(s) < 0$).
Marginally stable	Natural response oscillates indefinitely without growing or decaying. Poles on the imaginary axis.
Unstable	Natural response grows without bound. System can no longer be controlled. Poles in the right half s-plane ($\text{Re}(s) > 0$).

8. The Design Process

Designing a feedback control system follows an orderly sequence. The process is **iterative** — if performance is not met after analysis or simulation, the designer returns to an earlier step.

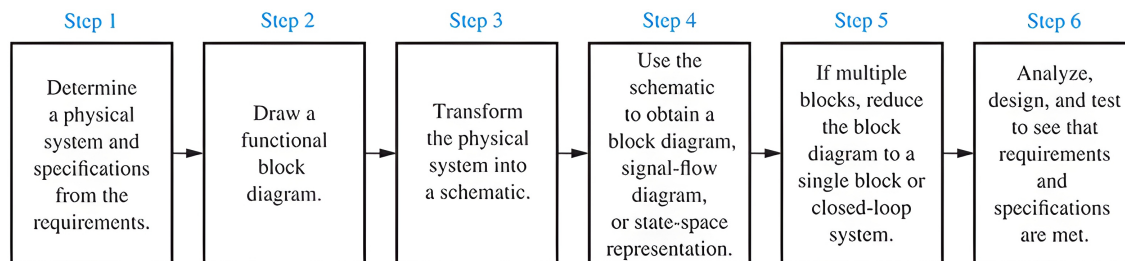


Figure 5: The control system design process flowchart

Step 1. Define system requirements

Identify specifications: transient response requirements, steady-state error limits, and stability requirements.

Step 2. Model the system

Derive the mathematical model — transfer function $G(s)$ or state-space representation (A, B, C, D) .

Step 3. Design the controller

Select the control strategy: PID, root locus compensation, or frequency domain methods.

Step 4. Analyze performance

Check stability, verify transient response specs, and compute steady-state error analytically.

Step 5. Simulate

Verify performance with MATLAB/Simulink before physical implementation. Adjust if needed.

Step 6. Implement and test

Build the prototype, test on hardware, and refine until all specifications are met.

Why is the process iterative?

Real systems rarely work perfectly on the first design attempt. Simulation may reveal stability issues not apparent in the analytical model; hardware testing may reveal modeling errors or unmodeled dynamics. Each iteration improves the design until all specifications are simultaneously satisfied.

Glossary of Key Terms

Term	Definition
Plant	The physical system or process being controlled, such as a motor, furnace, elevator, or robot arm.
Controller	The device or algorithm that computes the control signal based on the error or reference input.
Reference Input	The desired output value, also called the setpoint or command signal. Symbol: $r(t)$ or $R(s)$.
Error Signal	The difference between the reference input and the measured output: $e(t) = r(t) - y(t)$.
Feedback	The process of routing the measured output back to the summing junction for comparison with the reference.
Output Transducer	A sensor that measures the actual output and converts it to the form used by the controller.
Input Transducer	A device that converts the reference input to the form used by the controller.
Actuating Signal	The signal that drives the plant after the error has been processed by the controller.
Summing Junction	The point in the block diagram where the reference input and feedback signal are algebraically combined to produce the error signal.
Transfer Function	The Laplace-domain ratio of output to input for a linear, time-invariant system at rest: $G(s) = Y(s)/U(s)$.
Disturbance	An unwanted external signal that affects the plant output, such as wind load, friction, or road irregularities.
Transient Response	The behavior of a system from its initial state until it reaches its final steady-state value after an input is applied.
Steady-State Error	The difference between the desired output and the actual output after all transients have died out: $e_{ss} = \lim_{t \rightarrow \infty} e(t)$.
Natural Response	The response of the system due to its own dynamics, independent of the input. Determines stability.
Forced Response	The response of the system due to the applied input signal.
Stability	A system is stable if its natural response decays to zero (or remains bounded) as time approaches infinity.
Characteristic Equation	The equation $1 + G(s)H(s) = 0$, whose roots (the closed-loop poles) determine the stability of the system.
Closed-Loop Poles	The roots of the characteristic equation. Must have negative real parts for the system to be stable.
Open-Loop System	A control system in which the output has no effect on the input. No feedback path exists.
Closed-Loop System	A control system that uses feedback to compare the actual output with the desired output and drives the error to zero.